# PHYS 4041/5041:

# Mathematical Methods of Physics 1

# Differential Equations II

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# Differential Equations II

- Wronskian
- Series solution of ODEs
- Method of Frobenius
- Fuch's theorem

# Wronskian

- The Wronskian is discussed in Arfken, Weber, and Harris, Section 7.6.
- Consider two functions of $x$, $y_1$ and $y_2$.
- We define the *Wronskian* to be

$$W(x) = y_1 y_2' - y_1' y_2.$$

  If $y_1$ and $y_2$ are linearly dependent, then $W = 0$ everywhere.

- If $y_1$ and $y_2$ are *analytic* within an interval, then $W = 0$ in that interval implies that they are linearly dependent (converse).

# Finding second solutions

- Consider $y'' + p(x)y' + q(x)y = 0$.
- Using the ODE, we have $W' = -pW$. Therefore, $W = A\exp(-\int p\,dx)$, where $A$ is arbitrary.
  Side note: this implies that the Wronskian is a property of the ODE, rather than of $y_1$ and $y_2$.
- Now suppose we know $y_1$, and want to determine $y_2$. The definition of the Wronskian, $W = y_1y_2' - y_1'y_2$, provides a first-order ODE for $y_2$.
- This ODE may be solved by the method of an integrating factor to yield

$$y_2 = By_1 \int \frac{\exp(-\int p\,dx)}{y_1^2}\,dx.$$

# Series solution of ODEs

- See Boas, Sections 12.1, 12.11, and 12.21. The topic is covered in greater depth by Arfken, Weber, and Harris, Sections 7.4 and 7.5.

- We will limit ourselves to the second-order, linear, and homogeneous case: $y'' + p(x)y' + q(x)y = 0$.

- Strategy: assume a power series of the form:

$$y = x^s \sum_{n=0}^{\infty} a_n x^n,$$

where $a_0 \neq 0$. We have chosen a series around $x = 0$, which is typically what one wants to do. You may, however, choose to expand around an arbitrary $x = x_0$.

# Series solution of ODEs, continued

- Then substitute this series into the ODE, performing derivatives as needed. Then collect the terms into a single new power series.
- Since the differential equation is an identity, and power series are unique, these coefficients must vanish *term by term*.
- The requirement that $a_0 \neq 0$ will give a quadratic equation for $s$, known as the *indicial equation*.
- This approach is called the *Method of Frobenius* and is very general.
- One often finds recurrence formulas for the $a_n$. The convergence conditions may be checked using the ratio test.

# Taking derivatives

- Suppose we have:

$$y = \sum_{n=0}^{\infty} a_n x^{s+n}.$$

- We can then take derivatives:

$$y' = \sum_{n=0}^{\infty} (s+n) a_n x^{s+n-1}$$

and

$$y'' = \sum_{n=0}^{\infty} (s+n)(s+n-1) a_n x^{s+n-2}.$$

- Be aware that that $(x^0)' = 0$, not $x^{-1}$ !

# Re-labeling series

- In order to write the ODE is a single power series, it is usually necessary to re-label some of the terms.
- Suppose we have

$$y' = \sum_{n=0}^{\infty} (s+n) a_n x^{s+n-1} \qquad \text{and}$$

$$y'' = \sum_{n=0}^{\infty} (s+n)(s+n-1) a_n x^{s+n-2}.$$

- Then

$$y' = \sum_{\ell=-1}^{\infty} (s+\ell+1) a_{\ell+1} x^{s+\ell} \qquad \text{and}$$

$$y'' = \sum_{m=-2}^{\infty} (s+m+2)(s+m+1) a_{m+2} x^{s+m}.$$

# Analytic functions

- A function $f$ is said to be *analytic* at the point $x_0$ if it can be expressed as a Taylor series around $x = x_0$, and that series converges for some finite interval.

- Examples:
  - $f(x) = x^2 + 2x + 1$ is analytic everywhere.
  - $f(x) = 1/x$ is analytic everywhere, except $x = 0$.
  - $f(x) = \exp(-1/x^2)$ is analytic everywhere, except $x = 0$. What does this function and its power series look like?

# Singular points

- The point $x = x_0$ is said to be an *ordinary point* if $p(x)$ and $q(x)$ are both analytic at $x_0$.
- The point $x = x_0$ is said to be a *regular singular point* if either $p(x)$ or $q(x)$ is not analytic at $x_0$, but both $(x - x_0)p(x)$ and $(x - x_0)^2 q(x)$ are analytic at $x_0$.
- The point $x = x_0$ is said to be an *irregular singular point* if either $(x - x_0)p(x)$ or $(x - x_0)^2 q(x)$ are not analytic at $x_0$.
- The point at $x_0 \to \infty$ can be classified by changing variables to $u = x^{-1}$ and analyzing the resulting ODE at $u = 0$.

# Fuch's Theorem

- Suppose the $(x - x_0)p(x)$ and $(x - x_0)^2 q(x)$ are analytic around $x = x_0$, i.e., $x = x_0$ is an ordinary point or regular singular point (Fuchsian conditions). Then we are guaranteed to get at least one solution by the Frobenius method.

- This result is known as *Fuch's Theorem*.

- A guaranteed solution will correspond to the largest root of the indicial equation. It is guaranteed to converge for $|x - x_0| < R$, where $R$ is the distance from $x_0$ to the nearest singular point. $R$ will also be the smallest radius of convergence of $(x - x_0)p(x)$ and $(x - x_0)^2 q(x)$.

- One of the failure modes of the second solution is that it may not be linearly independent of the first solution.

- This implies that all bets are off, if $x = x_0$ is an irregular singular point.

# Fuch's Theorem, continued

More specifically: Fuchsian conditions are necessary and sufficient for the general solution of the ODE to be either

- two Frobenius series, or
- One Frobenius series $S_1(x)$, with the second solution being $S_1(x)\log x + S_2(x)$, where $S_2(x)$ is another Frobenius series. This is the case when the roots of the indicial equation or equal, and is often the case if the roots differ by an integer.

# More on convergence

- If the series is infinite, the convergence can be analyzed via the ratio test and/or by finding the singular points.
- There is another important possibility: the Frobenius series may terminate.
- This is the case for the power solutions to Euler's ODE and the Legendre, Chebyshev, and Laguerre polynomials, which are important in physical applications. For the special polynomials, the series only terminate for particular values of a parameter in the ODE.

# Hypergeometric Function

- For $|x| < 1$, the hypergeometric function is given by the Gauss series

$$
\begin{aligned}
{}_2F_0(a,b,c;x) &= \sum_{n=0}^{\infty} a_n x^n \\
&= 1 + \frac{ab}{c}\frac{x}{1!} + \frac{a(a+1)b(b+1)}{c(c+1)}\frac{x^2}{2!} + \dots
\end{aligned}
$$

- $a_0 = 1$, $a_{n+1} = \frac{(a+n)(b+n)}{(n+1)(c+n)} a_n$
- Note

$$
\frac{d}{dx}{}_2F_0(a,b,c;x) = \frac{ab}{c}{}_2F_0(a+1,b+1,c+1;x)
$$